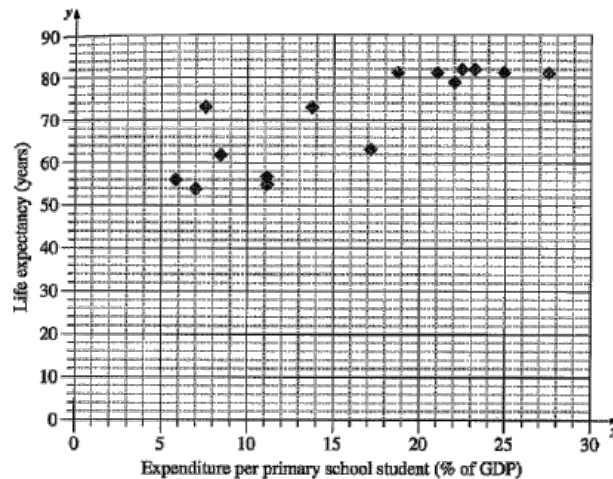


Bivariate Data Analysis Topic Test I

1. The scatterplot shows the relationship between expenditure per primary school student, as a percentage of a country's Gross Domestic Product (*GDP*), and the average life expectancy of the population, in years, for 15 countries.



- For the given data, the correlation coefficient, r , is 0.83. What does this indicate about the relationship between expenditure per primary school student and life expectancy for the 15 countries.
- For the data representing expenditure per primary school student, Q_L is 8.4 and Q_U is 22.5. What is the interquartile range?
- Another country has an expenditure per primary school student of 47.6% of its *GDP*. Would this country be an outlier for this set of data? Justify your answer with calculations.
- The expenditures per primary school student for the 15 countries in the scatterplot are:

5.9, 7, 7.6, 8.4, 11.2, 11.2, 13.7, 17.1, 18.7, 21.1, 22, 22.5, 23.2, 24.9, 27.9

Complete the table below by calculating the mean, $\bar{x}$, and the standard deviation, S_x , of these data. Calculate both values to two decimal places. The table also shows the mean, $\bar{y}$, and the standard deviation, S_y , of life expectancy for the same 15 countries.

	Mean	Standard deviation
Expenditure per primary school student	$\bar{x} =$	$S_x =$
Life expectancy	$\bar{y} = 70.73$	$S_y = 10.94$

- Using the values from the table in part (d), show that the equation of the least squares line of best fit is $y = 1.29x + 49.9$.

1(a)

Strong positive correlation because r is close to 1.

1(b)

$$\begin{aligned} \text{IQR} &= Q_3 - Q_1 \\ &= 22.5 - 8.4 = 14.1 \end{aligned}$$

1(c)

$$\begin{aligned} Q_3 + 1.5 \times \text{IQR} &= 22.5 + 1.5 \times 14.1 \\ &= 43.65 \end{aligned}$$

Now $47.6 > 43.65$

$\therefore$ This country would be an outlier.

1(d)

Use the statistics mode on calculator.

$$\bar{x} = 16.14 \text{ and } S_x = 7.03$$

1(e)

$$m = r \frac{S_y}{S_x} = 0.83 \times \frac{10.94}{7.037} = 1.29$$

$$\begin{aligned} b &= \bar{y} - m\bar{x} \\ &= 70.73 - 1.29 \times 16.14 = 49.9 \end{aligned}$$

$\therefore$ Equation of line of best fit is

$$y = 1.29x + 49.9$$

2. What is the y - intercept of the least-squares regression line given $m = 0.5$, $\bar{x} = 41.7$ and $\bar{y} = 58.4$?

$$\begin{aligned} c &= \bar{y} - m\bar{x} \\ &= 58.4 - 0.5 \times 41.7 = 37.55 \end{aligned}$$



Bivariate Data Analysis Topic Test I

3. A strong correlation exists between white bread intake at meals and increased glucose and insulin levels among people with type-2 diabetes.
- Is the correlational coefficient for this relationship positive or negative?
 - Does the statement suggest that white bread directly affects glucose and insulin in the body? Explain.

5(a)

Positive correlation. Increasing white bread increases the amount of glucose and insulin in the body.

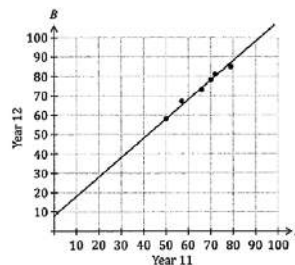
5(b)

No. The statement states there is a correlation, not causality.

4. The table below shows the weight of six students in Year 11 and their weight in Year 12.

Year 11 (kg)	50	57	66	70	72	79
Year 12 (kg)	58	67	73	78	81	85

Draw a scatterplot and a line of best fit for this data using Year 11 as the horizontal axis and Year 12 as the vertical axis.



5. What is the slope of the least-squares regression line given $r = 0.642$, $S_x = 2.567$ and $S_y = 5.123$?

$$m = r \frac{S_y}{S_x} = 0.642 \times \frac{5.123}{2.567} = 1.28$$

6. Suppose that the least squares regression line, which would enable the amount of money spent at the shopping centres (in dollars), to be predicted from the number of hours spent at the shopping centres is given by:

$$\text{Expenditure} = 30 + 75 \times \text{Hours}$$

- Predict the expenditure on shopping by an individual who spends 3 hours at the shopping centre.
- Predict the hours spent shopping by an individual whose shopping expenditure is \$1425.
- How would you interpret the slope of the line in this equation?

10(a)

$$\begin{aligned} \text{Expenditure} &= 30 + 75 \times \text{Hours} \\ &= 30 + 75 \times 3 \\ &= \$255 \end{aligned}$$

10(b)

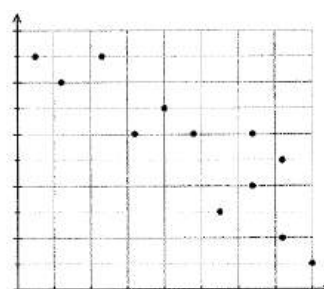
$$\begin{aligned} \text{Expenditure} &= 30 + 75 \times \text{Hours} \\ 1425 &= 30 + 75 \times \text{Hours} \\ 1425 &= 75 \times \text{Hours} \\ \text{Hours} &= \frac{1425}{75} = 18.6 \end{aligned}$$

10(c)

Slope of the line: on average, for each extra hour spent shopping, an extra \$75 is spent.

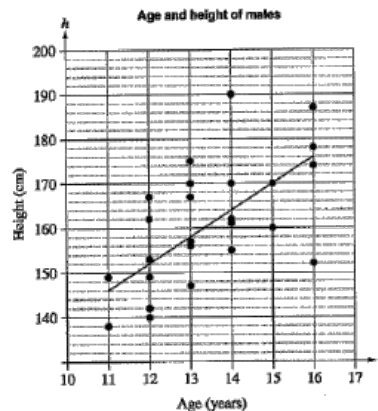
7. The correlation coefficient for two quantities was -0.5 . Draw a scatterplot that could represent this result.

Correlation coefficient of -0.5 is a negative linear relationship with medium strength.



Bivariate Data Analysis Topic Test I

8. Ahmed collected data on the age (a) and height (h) of males aged 11 to 16 years. He created a scatterplot of the data and constructed a line of best fit to model the relationship between the age and height of males.



- Determine the gradient of the line of best fit shown on the graph.
- Explain the meaning of the gradient in the context of the data.
- Determine the equation of the line of best fit shown on the graph.
- Use the line of best fit to predict the height of a typical 17-year-old male.
- Why would this model not be useful for predicting the height of a typical 45-year-old male?

12(a)

$$m = \frac{\text{Rise}}{\text{Run}} = \frac{176 - 146}{5} = 6$$

12(b)

Each year the height of males aged 11 to 16 years increases, on average, by 6 cm.

12(c)

Gradient-intercept formula

$$y = mx + c$$

$$h = 6a + c$$

To find the vertical-intercept, select a point on the line such as (11, 146).

$$h = 6a + c$$

$$146 = 6 \times 11 + c$$

$$b = 80$$

Equation of the line is $h = 6a + 80$.

12(d)

$$h = 6a + 80$$

$$= 6 \times 17 + 80 = 182 \text{ cm}$$

A typical 17-year-old male is 182 cm.

12(e)

Once a person reaches adulthood their height does not increase. Using the model for a 45-year-old male results in an inaccurate height of 350 cm.

9. The age, in years, for a sample of cars and their advertised selling price are given in the following table.

Age	2	3	4	4	5	5	6	6	6	7	7	8
Price (in \$1000)	31.6	28.6	26.0	23.6	22.0	24.4	19.6	19.0	20.0	19.4	16.6	16.0

Determine the equation of the least squares line that would allow the selling price of a second-hand car to be predicted from its age. Give the coefficients correct to four significant figures.

$$y = mx + c$$

$$m = r \times \frac{s_y}{s_x}, \quad c = \bar{y} - m\bar{x}$$

$$r = -0.9692$$

$$s_y = 4767.95$$

$$s_x = 1.7645$$

$$\bar{y} = 2233.33$$

$$\bar{x} = 5.25$$

$$m = -0.9692 \times \frac{4767.95}{1.7645} = -2619$$

$$c = 2233.33 - (-2619) \times 5.25 = 35980$$

$$\text{Price} = -2619 \times \text{Age} + 35980$$



Bivariate Data Analysis Topic Test I

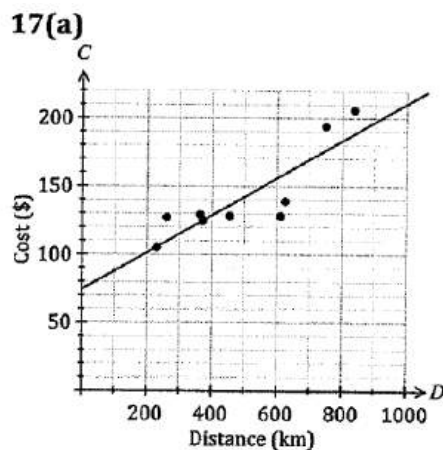
10. What is the y – intercept of the least-squares regression line given $m = 1.2$, $\bar{x} = 48$ and $\bar{y} = 71$?

$$\begin{aligned} c &= \bar{y} - m\bar{x} \\ &= 71 - 1.2 \times 48 \\ &= 13.4 \end{aligned}$$

11. A certain airline runs passenger flights from Sydney to various towns in Australia. The table below shows a sample of these flights, together with the distance of each flight and the cost for a passenger to take each flight.

Distance (D) km	Cost (C) dollars
231	105
260	127
364	129
371	125
455	128
611	128
625	139
750	194
837	206

- Construct a scatterplot below using the above table.
- Find Person's correlation coefficient. Answer correct to four decimal places.
- Determine the equation of the least-squares regression line. Answer correct to two decimal places.
- A flight from Sydney to Broken Hill takes 1014 km, and the airline charges passengers \$237 for this journey. Use your answer from (c) to determine whether this cost is too high, too low or about right.



17(b)
 $r = 0.86660754\dots$
 ≈ 0.8666

17(c)
 $A = 73.8587371\dots \approx 73.86$
 $B = 0.136827568\dots \approx 0.14$
 $y = mx + c$
 $y = Bx + A$
 $C = 0.14D + 73.86$

17(d)
 $C = 0.14 \times 1014 + 73.86$
 $= \$215.82$
 Therefore, the cost is too high.

12. The equation of least-squares line of best fit is given by $y = mx + c$ where $m = r \frac{S_y}{S_x}$ and $c = \bar{y} - m\bar{x}$. What is the gradient of the least-squares line of best fit given $r = 0.561$, $S_x = 1.987$ and $S_y = 4.579$?

- A. 0.24 B. 1.29 C. 7.12 D. 16.21

18 (B)
 $m = r \frac{S_y}{S_x}$
 $= 0.561 \times \frac{4.579}{1.987}$
 $= 1.29$

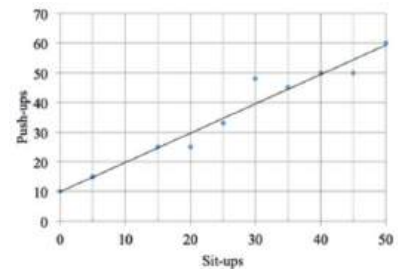
Bivariate Data Analysis Topic Test I

13. The Person's correlation coefficient between students' assessment result and their height was 0.12. What is the meaning of this correlation?

Assessment results increase as height increases.
Low positive correlation.
Not a strong relationship.

14. The scatterplot and line of best fit shows the number of sit-ups (s) and the number of push-ups (p) performed by students during a fitness test. Which of the following is the best estimate for Pearson's correlation coefficient for this data?

- A. -0.9
B. -0.5
C. 0.5
D. 0.9



15. Consider the scatterplot below. What is the correlation between the variables in the scatterplot?



- A. High positive
B. Low Positive
C. High negative
D. Low negative

16. The scatterplot shows the average number of children (per family) and the birth rate (per 100 000) for ten different countries. The line shown on the graph is the line of best fit for the data. The equation of the line is:

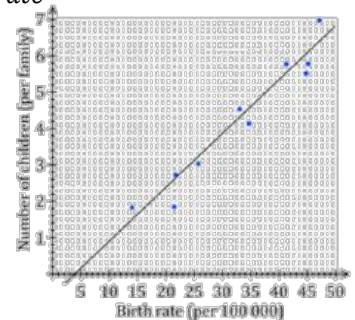
$$\text{average number of children} = -0.48 + 0.146 \times \text{birth rate}$$

Using this result, a prediction can be made for the average number of children (per family) for a country with a birth rate of 45 (per 100 000).

What is this prediction?

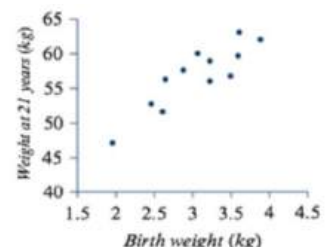
- A. 5.79
B. 6.09
C. 6.27
D. 6.57

$$\begin{aligned} \text{Average} &= -0.48 + 0.146 \times 45 \\ &= 6.09 \end{aligned}$$



17. The scatterplot shows the weights at age 21 and birth weight of 12 women. Describe the association in this scatterplot.

Strong positive linear



18. What is the gradient of the least-squares regression line given $r = 0.511$, $S_x = 0.365$ and $S_y = 9.461$?

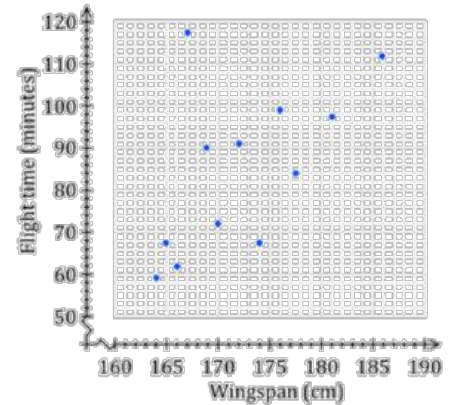
$$\begin{aligned} m &= r \frac{S_y}{S_x} \\ &= 0.511 \times \frac{9.461}{0.365} \\ m &= 13.2454 \end{aligned}$$



Bivariate Data Analysis Topic Test I

19. To investigate the association between flight time and wingspan for model aircraft to complete a certain flight, data were collected on 12 model aircraft. The scatterplot below was constructed.

The Pearson's correlation coefficient for this set of data is 0.56. If the outlying point (167, 117) is removed, which one of the following will be true for the correlation coefficient r for the remaining 11 model aircraft?



- A. $r > 0.56$ B. $r = 0.56$
C. $0 < r < 0.56$ D. $r < 0$

Since r represents the linear relationship between the wingspan and flight time, removing the outlier will strengthen this relationship so r will increase. $\therefore r > 0.56$

20. What is the best description between the steady increase in house price values and a Year 12 student's success in the HSC?

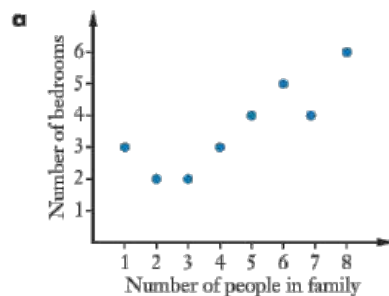
- A. Constant correlation B. Negative correlation
C. Positive correlation D. Zero correlation

Bivariate Data Analysis Topic Test 2

1. A group of 8 people were surveyed at random about how many people in their family lived at home. They were also asked about the number of bedrooms in their home.

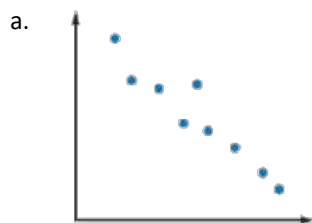
Number of people in family	4	3	6	7	2	1	5	8
Number of bedrooms	3	2	5	4	2	3	4	6

- Draw a scatterplot showing the data.
- Describe the shape of the scatterplot.
- Describe the relationship (if any) between the number of people in the family and the number of bedrooms in the home.



- The scatterplot is non-linear, or linear (positive) with outliers.
- With some exceptions, the more family members, the more bedrooms in the home.

2. Describe the pattern of each set of bivariate data. (linear or non-linear)



- Non-linear
- Linear in negative direction

3. What is Pearson's correlation coefficient (to 3 decimal places) for the table below?

x	22	23	24	25	26	27	28	29	30	31	32
y	32	29	27	24	18	17	14	11	8	4	1

using the calculator,

$$r = -0.997$$

4. The height and weight of eight men are given in the table below. What is the value of the Pearson's correlation coefficient?

Height (in cm)	170	172	175	176	178	180	183	184
Weight (in kg)	62.3	60.9	70.2	74.9	80.9	77.2	85.9	86.1

using the calculator,

$$r = 0.9575$$



Bivariate Data Analysis Topic Test 2

5. Which of the following has a positive correlation between the variables?

- A. Age when married and running ability.
- B. Number of police on the roads and the number of car breakdowns.
- C. Distance travelled and the cost of a taxi.
- D. The size of a person's foot and their intelligence.

6. The least squares regression line $y = -0.86x - 5.1$ predicts that when $x = 10$, what is the value of y ?

$$y = -0.86 \times 10 - 5.1$$
$$y = -13.7$$

7. The equation of a regression line that enables weekly amount spent on entertainment (in dollars) to be predicted from weekly income is given by:

$$\text{amount} = 0.10 \times \text{income} + 40$$

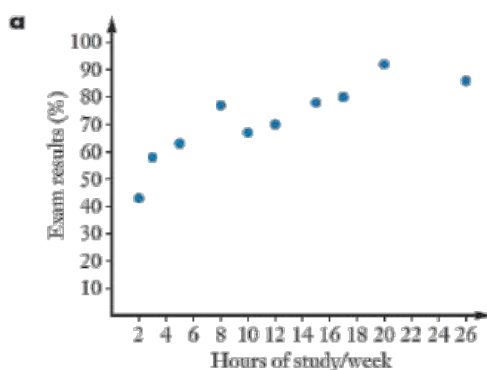
What is the predicted amount spent on entertainment with a weekly income of \$600?

$$\text{amount} = 0.1 \times 600 + 40$$
$$\text{amount} = \$100$$

8. The table shows the amount of study time and exam results for some students.

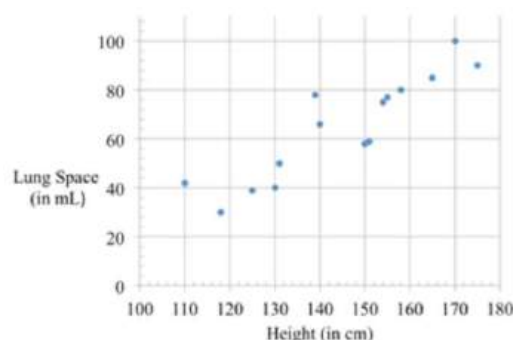
Hours of study/week	15	26	12	17	5	10	2	8	20	3
Exam results (%)	78	86	70	80	63	67	43	77	92	58

- a. Draw a scatterplot of the data.
- b. Describe the form of the scatterplot.
- c. What is the relationship between the amount of study and the exam results?



- b The shape is linear in a positive direction.
- c As the number of hours of study increases, exam results increase.

9. The scatterplot below shows the heights of the children and their lung space. What relationship is suggested by the scatterplot?



Strong positive correlation

Bivariate Data Analysis Topic Test 2

10. The cost of preparing meals in a school canteen is linearly related to the number of meals prepared. To help the caterers predict the costs, data were collected on the cost of preparing meals for different levels of demands. The data are shown below.

Number of meals	30	35	40	45	50	55	60	65	70	75	80
Cost (dollars)	138	154	159	182	198	198	214	208	238	234	244

What is the equation of the least-squares line of best fit?

$$y = mx + c$$

$$r = 0.9784$$

$$s_y = 35.5949$$

$$s_x = 16.5831$$

$$\bar{y} = 197$$

$$\bar{x} = 55$$

$$m = r \times \frac{s_y}{s_x} = 2.1$$

$$c = \bar{y} - m\bar{x} = 81.5$$

$$\therefore \text{Cost} = 2.1 \times \text{number of meals} + 81.5$$

11. Blood pressure levels for women increase as they get older. What is the best description for the relationship between blood pressure levels and a woman's age?

A. Constant correlation

B. Negative correlation

C. Positive correlation

D. Zero correlation

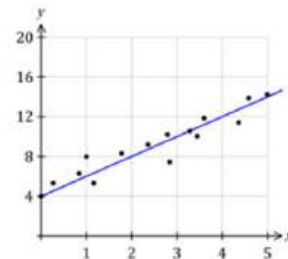
12. Find the equation of the line of best fit in this scatterplot.

$$y = mx + b$$

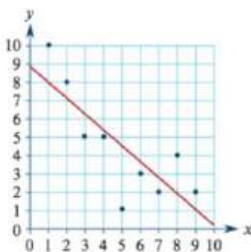
$$m = \frac{\text{rise}}{\text{run}} = \frac{8-4}{2} = 2$$

$$b = 4$$

$$\therefore y = 2x + 4$$



13. Find the equation of the least squares line plotted on the scatterplot.



$$y = mx + b$$

$$m = \frac{\text{rise}}{\text{run}} = -\frac{9}{10} = -0.9$$

$$b = 9$$

$$\therefore y = -0.9x + 9$$

14. The variables profit made and amount spent on advertising are strongly correlated with a correlation coefficient $r = 0.86$. From this information, it can be concluded that:

A. Increasing spending on advertising will decrease profits.

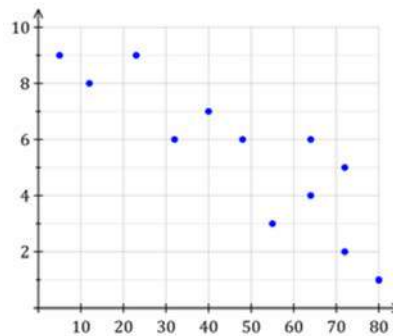
B. Increased spending of advertising is associated with increased profits.

C. The amount spent on advertising by a business is 86% of its profit.

D. The amount spent on advertising by a business is 74% of its profits.

Bivariate Data Analysis Topic Test 2

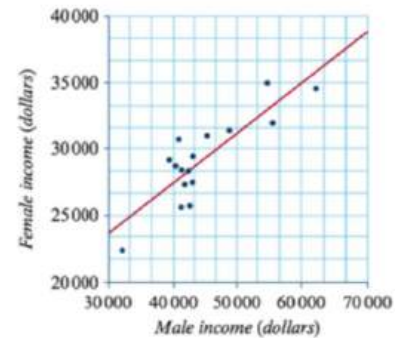
15. How would you describe the relationship in the scatterplot below?



A moderate negative correlation.

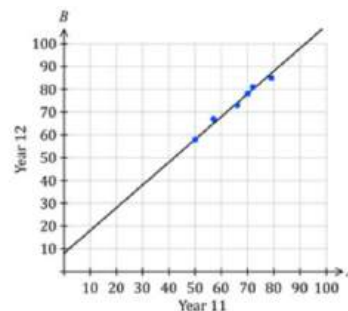
16. The scatterplot shows the average annual female income plotted against average annual male income for 16 countries. A least squares regression line is fitted to the data. What is the gradient of the least-squares line of best fit, correct to one decimal place?

- A. 0.9
- B. -0.6
- C. 0.4
- D. -0.4



17. What is the equation of the line of best fit in this scatterplot?

- A. $B = -A + 4$
- B. $B = -A + 8$
- C. $B = A + 4$
- D. $B = A + 8$



18. Determine whether each pair of variables are likely to have a causal relationship.

- a. Head circumference and weight
- b. Population of a city and pollution

a). No causal relationship
b). Causal relationship

Bivariate Data Analysis Topic Test 2

19. A block of ice was taken out of a freezer and left to thaw. The results are in the table below.

Time t (min)	5	10	15	20	25	30	35	40
Mass m (kg)	23.7	18.8	11.3	8.7	6.2	5.5	2.3	1.5

- Find the correlation coefficient. Is there a high correlation? Is it positive or negative? What does this mean?
- Find the equation of the least-squares regression line.
- Use the equation to estimate the mass of the ice after:
 - 18 minutes
 - an hour
- Discuss why extrapolation may not be useful in this situation.

a). $r = -0.954$, which shows a high negative correlation. This means that the regression line has a negative gradient and that the mass decreases as time increases.

b). $S_y = 7.8762$
 $S_x = 12.2474$
 $\bar{x} = 22.5$
 $\bar{y} = 4.75$
 $y = mx + c$
 $m = r \times \frac{S_y}{S_x} = -0.614$
 $c = \bar{y} - m\bar{x} = 23.565$

$\therefore m = -0.614t + 23.565$

c). (i). when $t = 18$:

$$m = -0.614 \times 18 + 23.565$$

$$m = 12.513 \text{ kg}$$

(ii). when $t = 60$:

$$m = -0.614 \times 60 + 23.565$$

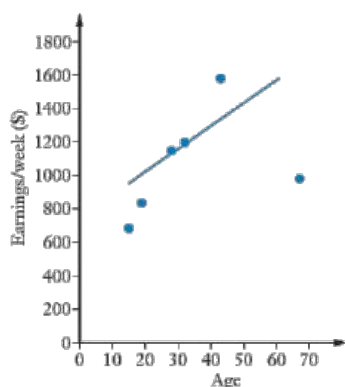
$$m = -13.275 \text{ kg}$$

d). The answer to (c)(ii). shows that extrapolation is not useful in this situation. The ice has melted to 0 volume before an hour has passed and the equation is no longer relevant.

20. The table shows the results of a survey into ages and earnings of a group of people.

Age	15	32	19	28	43	67
Earnings/week (\$)	689	1205	840	1154	1587	986

- Draw a scatterplot for this data.
- Find the correlation coefficient.
- Find the equation of the least-squares regression line.
- From this equation find the earnings of a 50-year-old.
- Is this equation a good model to extrapolate?



b). $r = 0.395$
 c). $S_y = 315.41$
 $S_x = 18.95$
 $\bar{x} = 34$
 $\bar{y} = 1076.83$

$$m = r \times \frac{S_y}{S_x} = 6.57$$

$$c = \bar{y} - m\bar{x} = 853.45$$

$\therefore y = 6.57x + 853.45$

d). when $x = 50$:

$$y = 6.57 \times 50 + 853.45$$

$$y = 1181.95$$

e). No it is not. For example, using the equation to find the age of a person earning \$1800 would give a 145-year-old!